

Zeynep S. Amac

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EDUCATION

Indiana University Bloomington

May 2029

Luddy School of Informatics, Computing, and Engineering

Bachelor of Science in Intelligent Systems Engineering (Bioengineering Concentration)

Luddy Living Learning Member | Hutton Honors Scholar | Provost Scholar | Groups Scholar

TECHNICAL SKILLS

Languages: Python, C, R

Developer Tools: Git/GitHub, Linux

Libraries & Visualization: Pandas, NetworkX, NumPy, Matplotlib, Vega, Jupyter Notebook, Gephi

Design & Creative Software: Photoshop, Illustrator, Premiere Pro, Lightroom, InDesign, Fusion 360, PrusaSlicer

WORK EXPERIENCE

Undergraduate Research Assistant

May 2026-Present

Laboratory for Computational Epidemiology and Public Health - Indiana University Bloomington

Bloomington, IN

- Apply bioengineering and data-analysis skills to support infectious-disease and mosquito-forecasting research, using quantitative models to inform public-health decision-making.

Undergraduate Research Assistant (IUUR)

Jul 2024-Present

Cyberinfrastructure for Network Science Center (CNS) -Indiana University Bloomington

- Updated and validated more than 4,000 anatomical structures for the Human Reference Atlas using Python and Jupyter Notebooks.
- Validated biological relationships and adapted the visualization pipeline to generate organ-specific multiscale trees.
- Developed and documented a Python utility that automated repetitive visualization-update tasks for future Human Reference Atlas releases.
- Presented research on Human Reference Atlas visualization at the IU 2025 Summer Poster Symposium.
- Initially served as a graphic designer, creating visual and video content to communicate CNS research and increasing social-media engagement by approximately 50%.
- Produced visual content for *Atlas of Macroscopes*, published by The MIT Press.

Teaching Assistant

Nov 2025-Present

Indiana University Bloomington - Innovation and Design

- Provide feedback, grade assignments, and mentor students during class and office hours for a class of 150.
- Guide students during class and office hours, reinforcing engineering and design concepts.

Peer Mentor

Aug 2026-Present

Indiana University Bloomington

- Support incoming engineering students through Innovation and Design instruction, tutoring, workshops, and mentoring events.

PROJECTS

Human Reference Atlas Organ-Specific Visualization | *Research Project*

May 2026-Present

- Built a Vega-based visualization pipeline for organ-specific multiscale anatomical trees.
- Created reusable Python workflows and configuration files for organ-specific visualization generation.
- Developed scientific visualizations for an upcoming book in Dr. Katy Börner's Atlas series.

Environmental Awareness Bee Game | *Software Systems Engineering Course*

Apr 2026-May 2026

- Implemented a grid-based educational game in C using arrays, structures, functions, and file handling.
- Promoted environmental awareness through interactive gameplay and strategic decision-making.

Robot Rescue | *Innovation and Design Course*

Nov 2025-Dec 2025

- Designed an educational board game teaching AI literacy and iterative learning to students ages 10-14.
- Presented the design process and completed prototype in a final course presentation to peers and instructors.

COMMUNITY INVOLVEMENT

Director of Marketing & Communications

Apr 2026-Present

ColorStack at IU

Indiana University Bloomington

- Manage chapter branding, social media, and event promotions across multiple platforms.

Member

Aug 2025-Present

Turkish Student Association at IU

Indiana University Bloomington

- Organize annual events for more than 300 attendees and recruited more than 40 new members through the Involvement Fair.

CONFERENCES

Grace Hopper Conference

Nov 2025

Conference Attendee

Chicago, IL

- Selected as one of 40 students from more than 200 applicants to attend the Grace Hopper Celebration.
- Connected with major companies, including Amazon, Disney, and Goldman Sachs, at a large career expo.